UNIT-3

Data link layer

Functionality of Data link Layer

- What is Data Link Layer?

The **Data Link layer** transforms a raw transmission facility into a line that appears free of undetected transmission errors to the network layer and is responsible for node-to-node delivery. It makes error-free the physical layer appear to the upper layer (network layer).

- **Data Link Layer** is the second layer of OSI Layered Model after Physical Layer.
- When a packet or message reaches to a network, it is the responsibility of **Data Link Layer** to transmit it to the Host using its MAC address.
- **Data Link Layer** devices are Switch & Bridges.

Sublayers of Data Link Layer

- Data Link Layer has two sublayers,

1. **Logical-Link-Control(LLC)**:
   This is the uppermost sub-layer, LLC consists of protocols running at the top of the **data link layer**, and also provides flow control, acknowledgment, and error notification. The LLC provides addressing and data link control. It specifies which methods are to be used for addressing channels over the transmission medium and for controlling the data exchanged between the generator of packet and recipient of the message.
2. **Media Access Control (MAC)**:
   Who can access the media at any one time, determines by the MAC sublayer(e.g. CSMA/CD).

The packet obtains from the Network layer is further divided into frames depending on the frame size of NIC(Network Interface Card). DLL also encases Sender's and Receiver's MAC address in the header.

An ARP(Address Resolution Protocol) request is placed onto the wire asking **"Who has that IP address?"** and after getting this message destination host replies with its MAC address. By this, the receiver's MAC address is obtained.

Design Issues with Data Link Layer

- The issue that arises in the **data link layer** (and most of the higher layers as well) is the speed mismatch between sender and receiver.
- Some traffic regulation mechanism is often needed to know the exact available free space, the receiver has at the moment. Frequently, the flow regulation and error handling are combined in this layer.
- Here is an additional issue in the **data link layer** with broadcast networks that is how to control access to the shared channel. The Medium Access Control (MAC), the sub-layer of the **data link layer** deals with this problem.

Functions of Data Link Layer

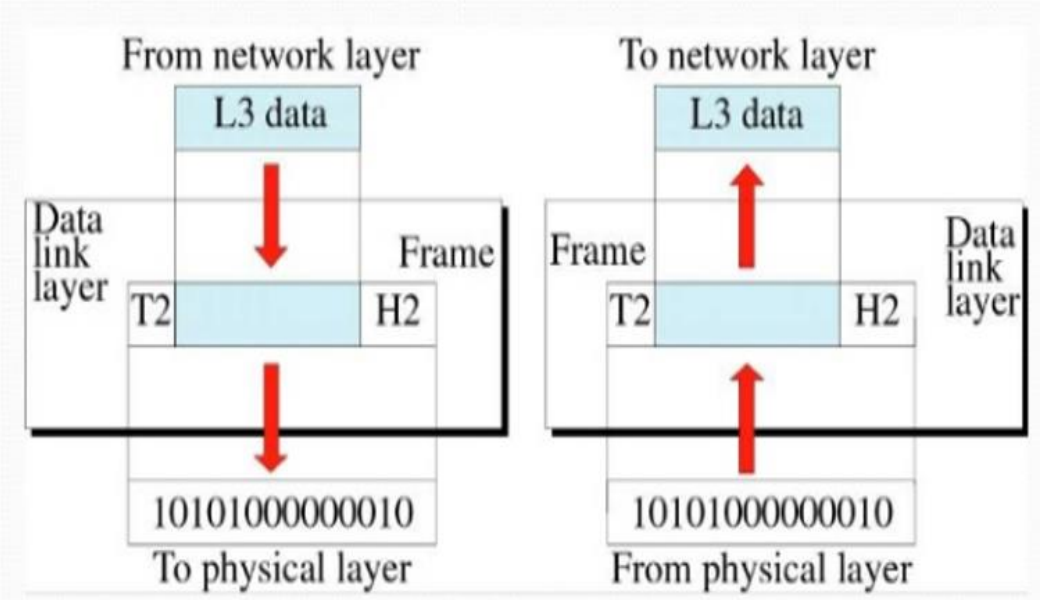
Specific functions of the **data link layer** include the following,

Framing: The **data link layer** receives the stream of bits from the network layer divides into manageable data units called frames.

Physical addressing: If frames are to be distributed to different stations on the network. To define the physical address of the sender (source address) and/or receiver (destination address) of the frame, the DLL adds a header to the frame.

If the frame is to be sent for a system outside the sender's network. The receiver's address means the address of the device that connects one network to the another.

This figure shows the relationship of the Data link layer to the network and physical layers.



Data Link Layer

Flow control: If the rate at which the data are consumed by the receiver is less than the rate produced by the sender, the data link layer deals with a flow control mechanism to prevent overrun the receiver.

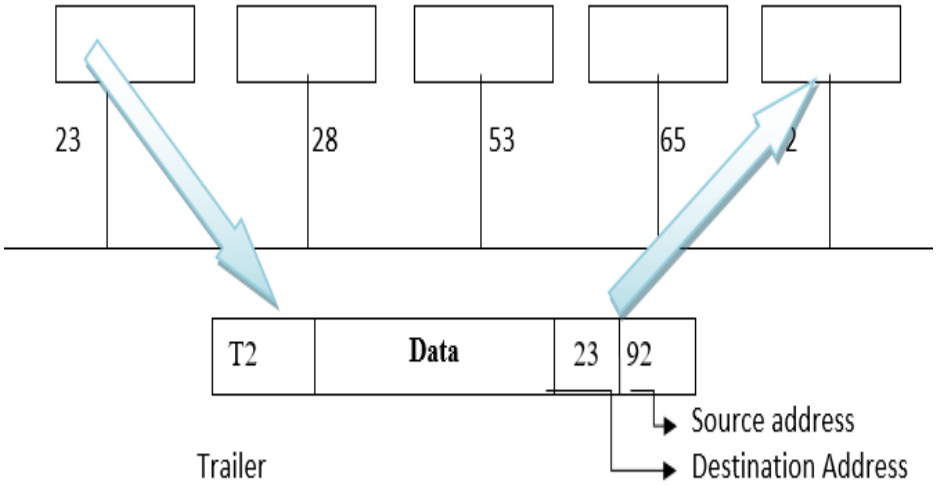
Error control: The data link layer also deals with damaged or lost frames. By adding mechanisms to detect and retransmit lost frames increases reliability.

A trailer added to the end of the frame to achieve error control.

Access control: When more than two or two devices are connected to the common link, data link layer protocols are necessary to determine which device has control over the link at any point of time.

Example: In this figure, a node with physical address 23 sends a frame to a node with physical address 92. The two nodes connected by a link level. At the link level this frame contains physical (link) addresses in the header.

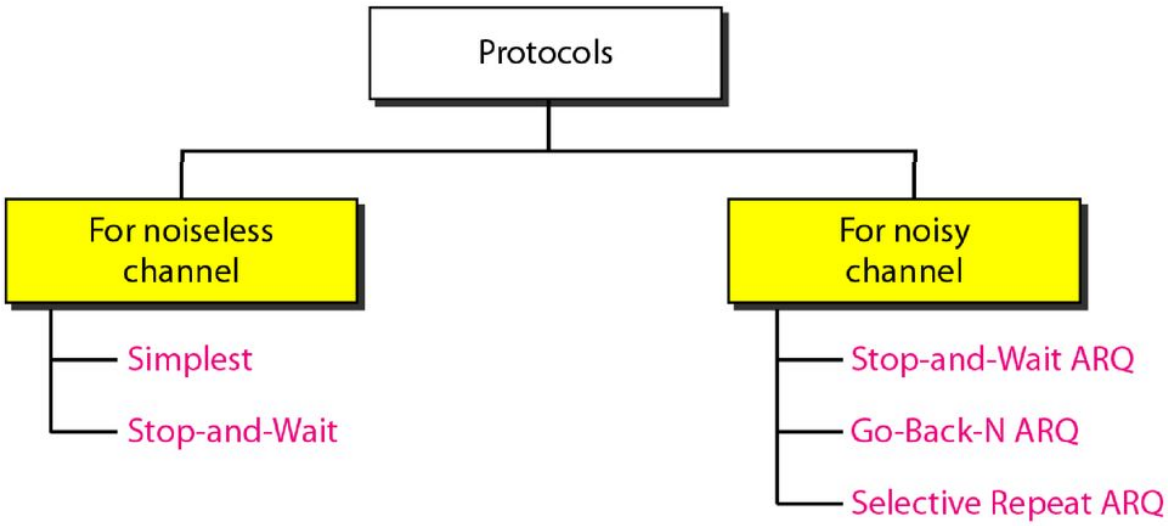
These are the only addresses needed. The remaining header contains other information that is required at this level. The extra bits needed for error detection contains by the trailer.



FLOW AND ERROR CONTROL

- Flow control refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgment.
- Error control in the data link layer is based on automatic repeat request, which is the retransmission of data.

PROTOCOLS



Noiseless Channel:

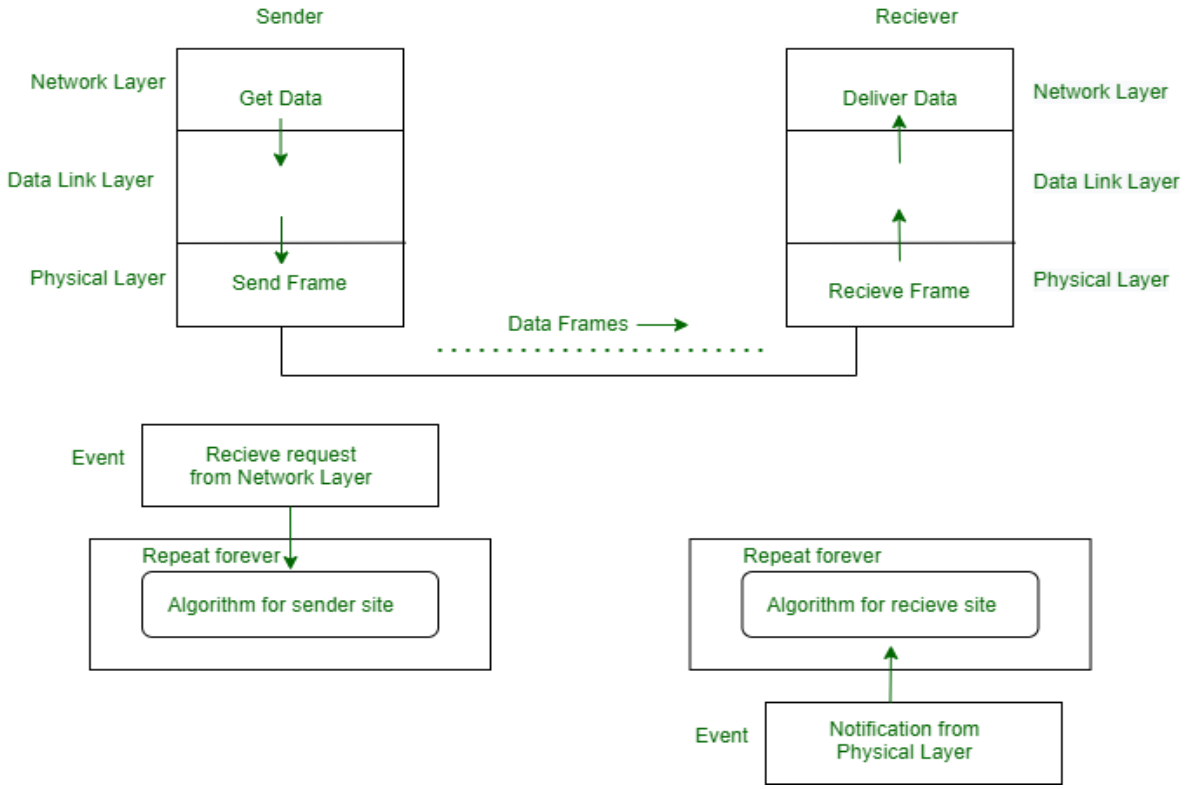
An idealistic channel in which no frames are lost, corrupted or duplicated. The protocol does not implement error control in this category. There are two protocols for the noiseless channel as follows.

Simplest Protocol –

We consider here that the receiver can maintain any frame received with insignificant processing time. The receiver's data link layer immediately removes the header from the frame and assigns the data packet to its network layer, which can also accept the packet immediately. That is to say, the receiver can never be overwhelmed with forthcoming frames.

- **Design:**

  The data link layer at the sender site gets data from its network layer, makes a frame out of the data, and sends it. The data link layer (receiver site) receives a frame from its physical layer, extracts data from the frame, and convey the data to its network layer. The data link layers of the sender and receiver provide communication/transmission services for their network layers. The data link layers utilization the services provided by their physical layers for the physical transmission of bits.



Design of Simplest Protocol with no error control or flow

- **Sender-site and Receivers algorithms :**
- **Sender-site algorithm –**

```
while (true)                          // Repeat forever
{
  WaitForEvent()i                     // Sleep until an event occurs
  if(Event(RequestToSend»             //There is a packet to send
  {
      GetData()i
      MakeFrame()i
      SendFrame()i                    //Send the frame
  }
}
```

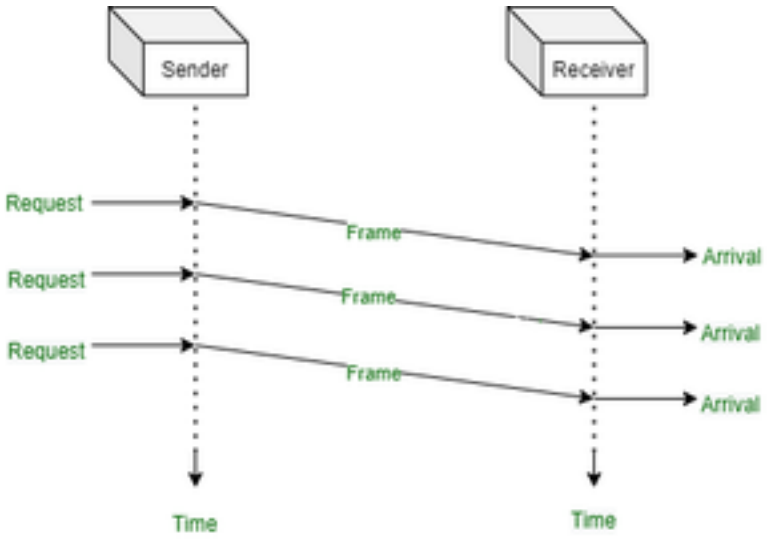
- **Receivers algorithm –**

```
while(true)                           // Repeat forever
{
  WaitForEvent()i                     II Sleep until an event occurs
  if(Event(ArrivalNotification»       IIData frame arrived
  {
      ReceiveFrame()i
      ExtractData()i
      DeliverData ()i                 //Deliver data to network layer
  }
}
```

- **Flow Diagram:**

  This Flow Diagram shows an example of communication using the simplest protocol. It is very straightforward. The sender sends a series of frames without further consideration about the receiver. Let's take an example, three frames will send from the sender, and three frames received by receivers. Bear in mind the data frames are shown by tilted boxes; the height of the box defines the transmission time difference between the first bit and the last bit in the frame.

- **Diagram** –



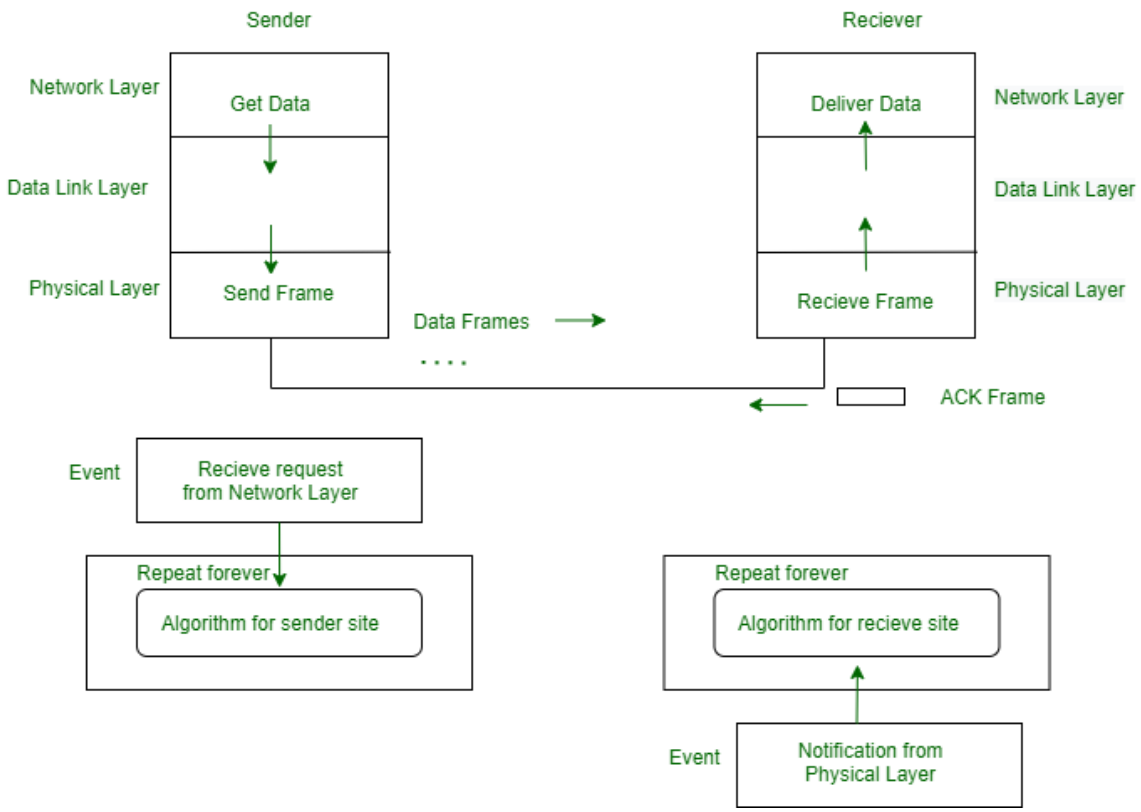
Flow Diagram for Simplest Protocol

Stop and Wait Protocol –

If data frame receivers arrive at the site faster than they can be processed, then frames must be stored until their use. Generally, the receiver does not have enough storage space, especially if it is receiving data from multiple sources.

- **Design:**

  On comparing the stop-and-wait protocol design model with the simplest protocol design model, we can see the traffic on the front/forward channel (from sender to receiver) and the back/reverse channel. Anytime, there is either one data frame on the forward channel or one ACK frame on the reverse channel. We hereupon require a half-duplex link.



Design of Stop-and-Wait Protocol

- **Sender-site and Receivers algorithms :**

  Sender-site algorithm –

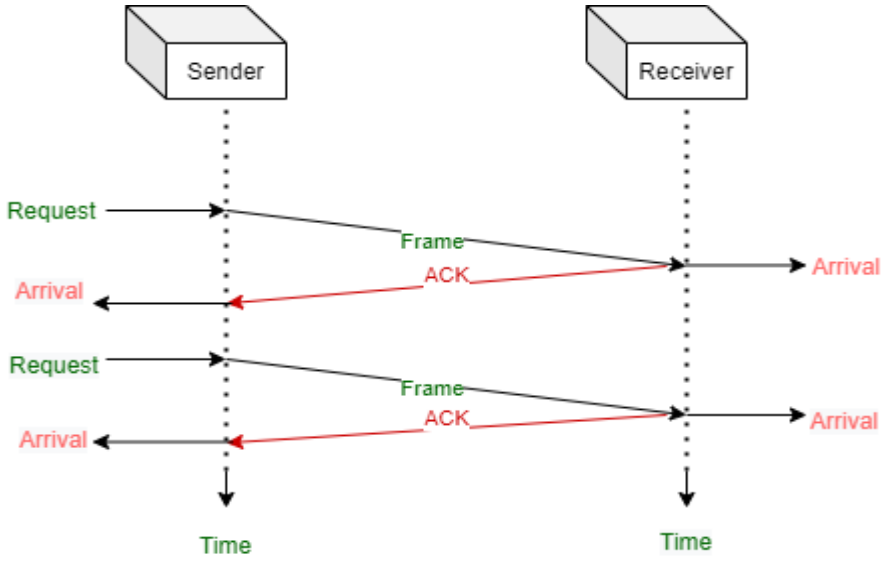
```
while(true)                          IIRepeat forever
canSend = true                       IIAllow the first frame to go
{
  WaitForEvent()i                    II Sleep until an event occurs
  if(Event(RequestToSend) AND canSend}
  {
     GetData()i
     MakeFrame();
     SendFrame()i                    I/Send the data frame
     canSend = false;                I/cannot send until ACK arrives
  }
  WaitForEvent()i                    II Sleep until an event occurs
  if(Event(ArrivalNotification)      /I An ACK has arrived
   {
     ReceiveFrame();                 I/Receive the ACK frame
     canSend = true;
   }
}
```

- **Receivers algorithm –**

```
while (true)                          IIRepeat forever
{
  WaitForEvent();                     II Sleep until an event occurs
  if(Event(ArrivalNotification)}      IIData frame arrives
  {
      ReceiveFrame(};
      ExtractData(}i
      Deliver(data};                  /IDeliver data to network layex
      SendFrame();                    IISend an ACK frame
  }
}
```

- **Flow Diagram:**

  This figure shows an example of communication using the Stop-and-wait protocol. It is still straightforward. The sender sends a frame and waits for a response from the receiver. When ACK(acknowledged) will arrival from the receiver side then send the next frame and so on. Keep it in mind when two frames will be there sender will involve in four event and the receivers will involve in two events.



Flow diagram for Stop & Wait Protocol

Error detection and Correction

- Data can be corrupted during transmission. Some applications require that errors be detected and corrected.

Types of Errors

Single-Bit Error

- The term *single-bit error* means that only 1 bit of a given data unit (such as a byte, character, or packet) is changed from 1 to 0 or from 0 to 1.
- **In** a single-bit error, only 1 bit in the data unit has changed.

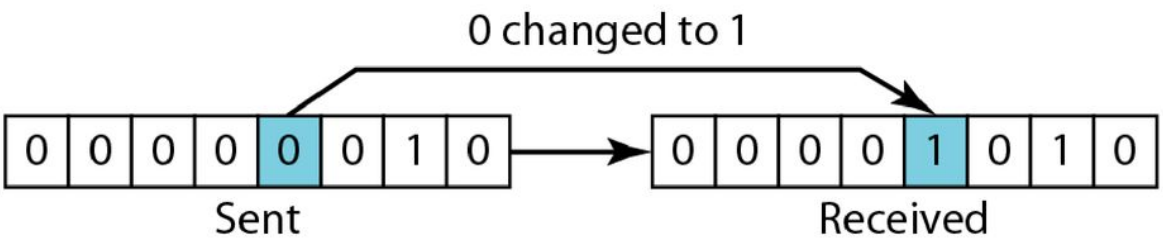


Figure: *Single-bit error*

- Single-bit errors are the least likely type of error in serial data transmission.

Burst Error

- The term *burst error* means that two or more bits in the data unit have changed from 1 to 0 or from 0 to 1.
- A burst error means that 2 or more bits in the data unit have changed.
- To understand why, imagine data sent at 1 Mbps. This means that each bit lasts only 1/1,000,000 s, or 1)ls.
- For a single-bit error to occur, the noise must have a duration of only 1)ls, which is very rare; noise normally lasts much longer than this.

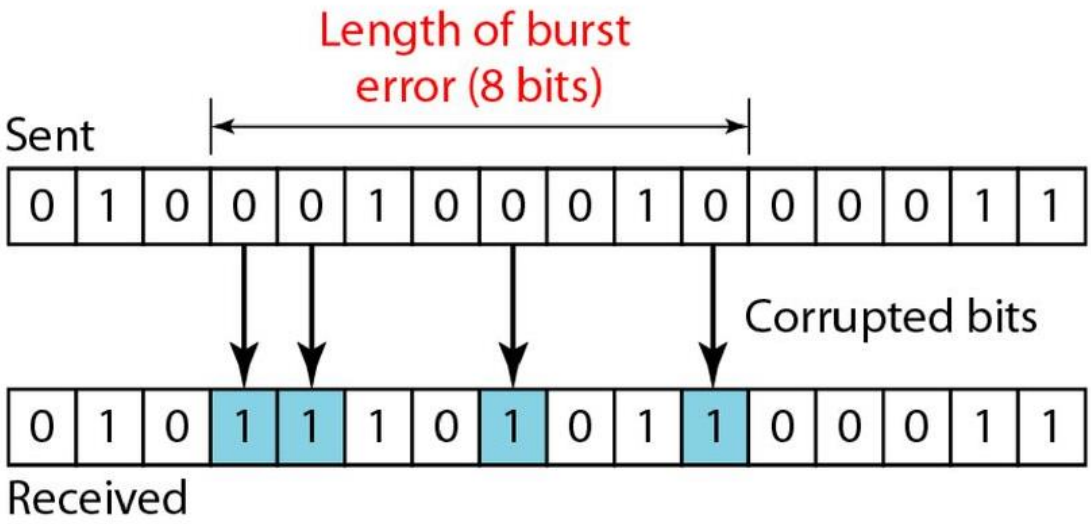


Figure: *Burst error of length 8*

- A burst error is more likely to occur than a single-bit error.
- The duration of noise is normally longer than the duration of 1 bit, which means that when noise affects data, it affects a set of bits.

Redundancy

- To detect or correct errors, we need to send extra (redundant) bits with data.

Detection versus Correction

- The correction of errors is more difficult than the detection. In error detection, we are looking only to see if any error has occurred.
- In error correction, we need to know the exact number of bits that are corrupted and more importantly, their location in the message.

Forward Error Correction versus Retransmission

- There are two main methods of error correction.
- **Forward error correction** is the process in which the receiver tries to guess the message by using redundant bits.
- This is possible, as we see later, if the number of errors is small.
- Correction by retransmission is a technique in which the receiver detects the occurrence of an error and asks the sender to resend the message.

- Resending is repeated until a message arrives that the receiver believes is error-free

Coding

- Redundancy is achieved through various coding schemes.
- The sender adds redundant bits through a process that creates a relationship between the redundant bits and the actual data bits.
- The receiver checks the relationships between the two sets of bits to detect or correct the errors

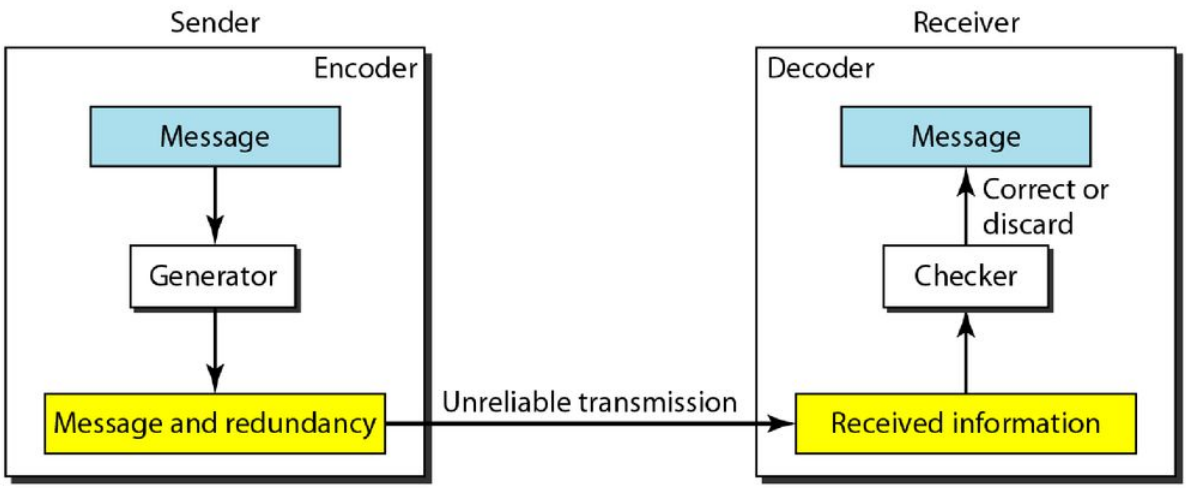


Figure: *The structure of encoder and decoder*

- We can divide coding schemes into two broad categories:
    - Block coding and Convolution coding.

Modular Arithmetic

- In modular arithmetic, we use only a limited range of integers.
- We define an upper limit, called a modulus *N*.
- We then use only the integers 0 to *N* - I, inclusive. This is *modulo-N* arithmetic.
- For example, if the modulus is 12, we use only the integers 0 to 11, inclusive

Modulo-2 Arithmetic

- In this arithmetic, the modulus *N* is 2.
- We can use only 0 and 1. In this arithmetic we use the XOR (exclusive OR) operation for both addition and subtraction.

- The result of an XOR operation is 0 if two bits are the same; the result is 1 if two bits are different.

Modulo-2 Arithmetic

- **Adding** 0+0=0 0+1=1 1+0=1 1+1=0
- **Subtracting** 0-0=0 0-1=1 1-0=1 1-1=0

Figure 10.4 *XORing of two single bits or two words*

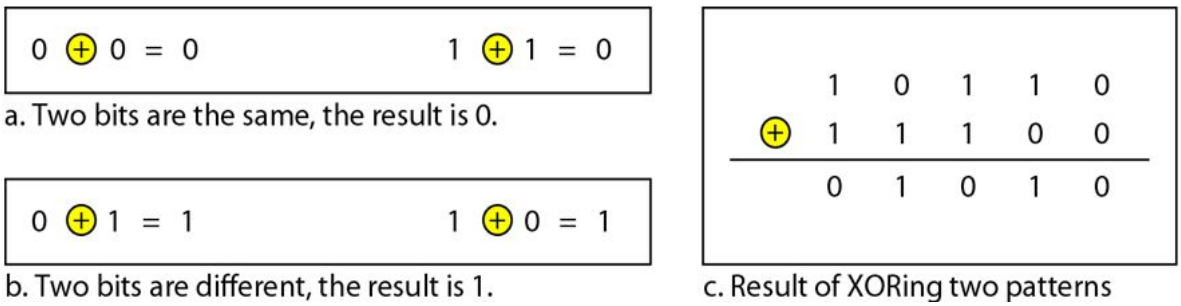


Figure: *XORing of two single bits or two words*

Introduction, Block Coding

- In block coding, we divide our message into blocks, each of *k* bits, called **datawords.**
- We add *r* redundant bits to each block to make the length $n = k + r$.
- The resulting *n-bit* blocks are called **codewords.**
- With *k* bits, we can create a combination of *2k* datawords; with *n* bits, we can create a combination of *2n* codewords.
- Since $n > k$, the number of possible codewords is larger than the number of possible datawords.

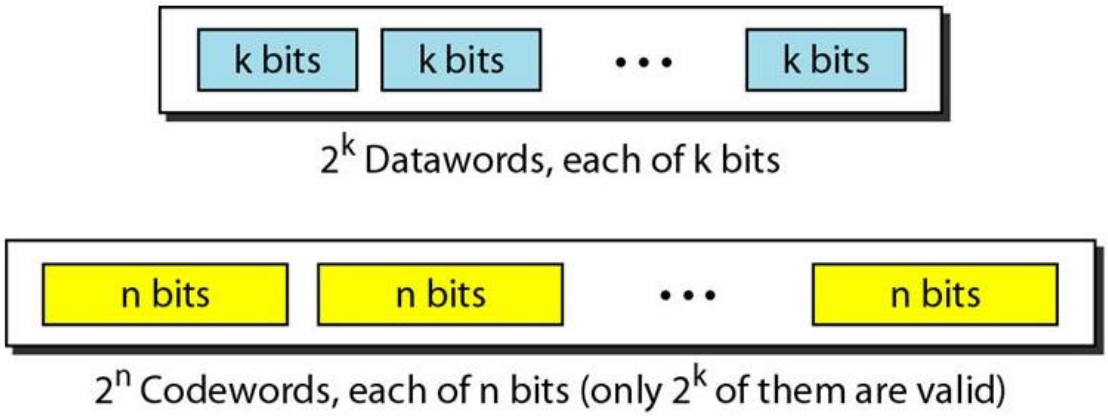


Figure: *Datawords and codewords in block coding*

Error Detection

- If the following two conditions are met, the receiver can detect a change in the original codeword.

  1. The receiver has (or can find) a list of valid codewords.
  2. The original codeword has changed to an invalid one.

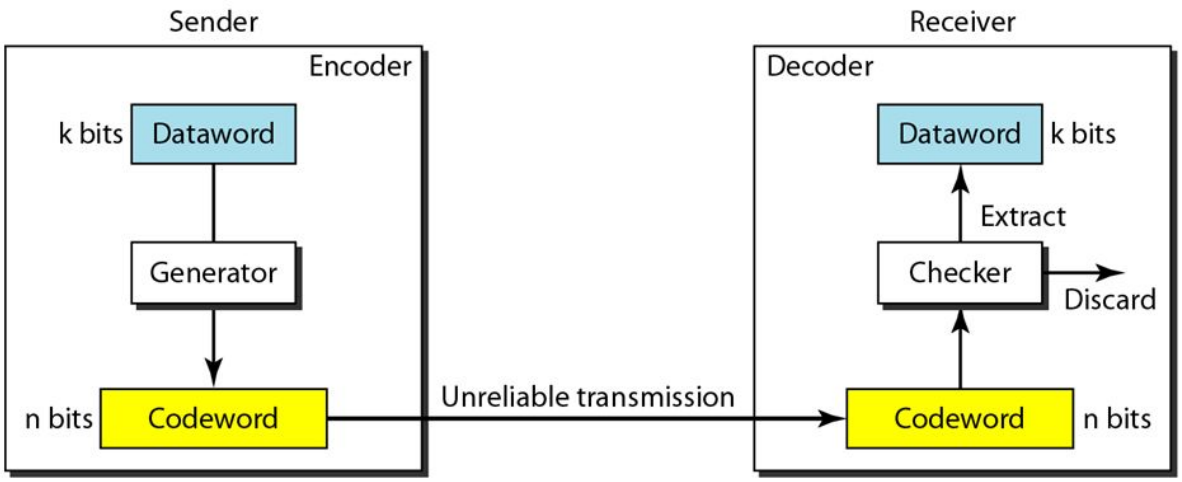


Figure: *Process of error detection in block coding*

Error Correction

- In error detection, the receiver needs to know only that the received codeword is invalid; in error correction the receiver needs to find (or guess) the original codeword sent.
- We can say that we need more redundant bits for error correction than for error detection.

- Figure shows the role of block coding in error correction.
- We can see that the idea is the same as error detection but the checker functions are much more complex.

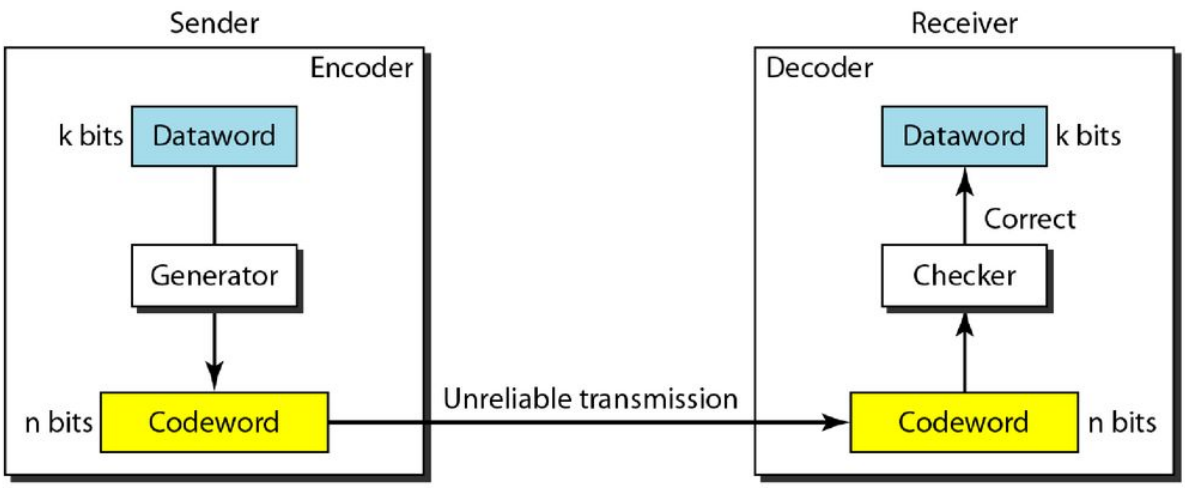


Figure: *Structure of encoder and decoder in error correction*

LINEAR BLOCK CODES

- In a linear block code, the exclusive OR (XOR) of any two valid codewords creates another valid codeword.

FRAMING

- Data transmission in the physical layer means moving bits in the form of a signal from the source to the destination.
- The data link layer, on the other hand, needs to pack bits into frames, so that each frame is distinguishable from another.
- Framing in the data link layer separates a message from one source to a destination,or from other messages to other destinations, by adding a sender address and a destination address.
- The destination address defines where the packet is to go; the sender address helps the recipient acknowledge the receipt.

Problems in Framing –

- **Detecting start of the frame:** When a frame is transmitted, every station must be able to detect it. Station detects frames by looking out for a special sequence of bits that marks the beginning of the frame i.e. SFD (Starting Frame Delimiter).

- **How does the station detect a frame:** Every station listens to link for SFD pattern through a sequential circuit. If SFD is detected, sequential circuit alerts station. Station checks destination address to accept or reject frame.
- **Detecting end of frame:** When to stop reading the frame.

Types of framing – There are two types of framing:

1. Fixed size – The frame is of fixed size and there is no need to provide boundaries to the frame, the length of the frame itself acts as a delimiter.

- **Drawback:** It suffers from internal fragmentation if the data size is less than the frame size
- **Solution:** Padding

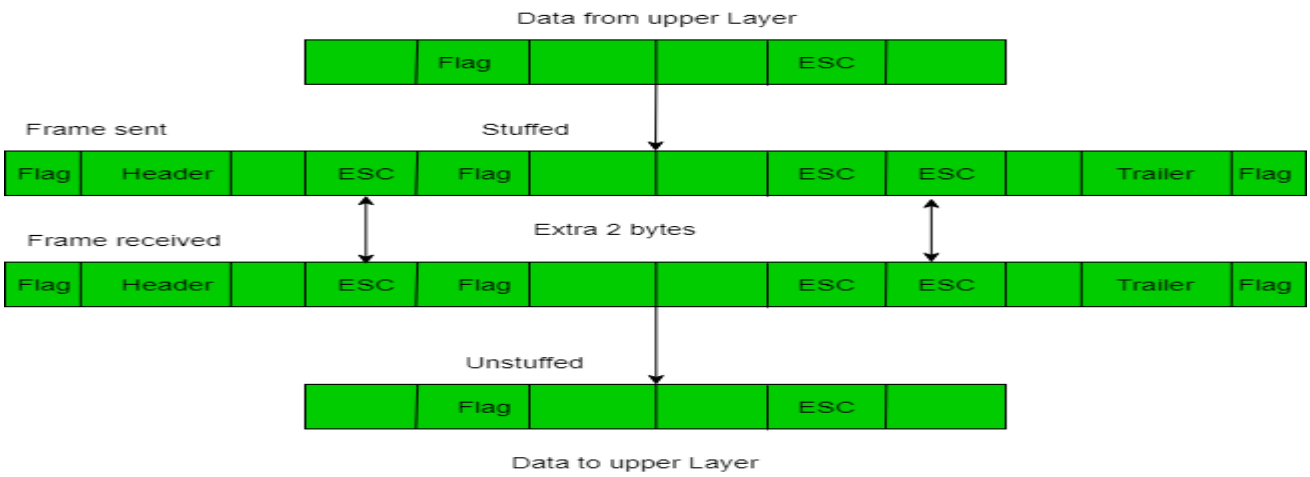
2. Variable size – In this, there is a need to define the end of the frame as well as the beginning of the next frame to distinguish. This can be done in two ways:

1. **Length field –** We can introduce a length field in the frame to indicate the length of the frame. Used in **Ethernet (802.3)**. The problem with this is that sometimes the length field might get corrupted.
2. **End Delimiter (ED) –** We can introduce an ED (pattern) to indicate the end of the frame. Used in **Token Ring**. The problem with this is that ED can occur in the data. This can be solved by:

   1. Character/Byte Stuffing: Used when frames consist of characters. If data contains ED then, a byte is stuffed into data to differentiate it from ED.

   Let ED = "$" –> if data contains '$' anywhere, it can be escaped using '\O' character.

   –> if data contains '\O$' then, use '\O\O\O$'($ is escaped using \O and \O is escaped using \O).



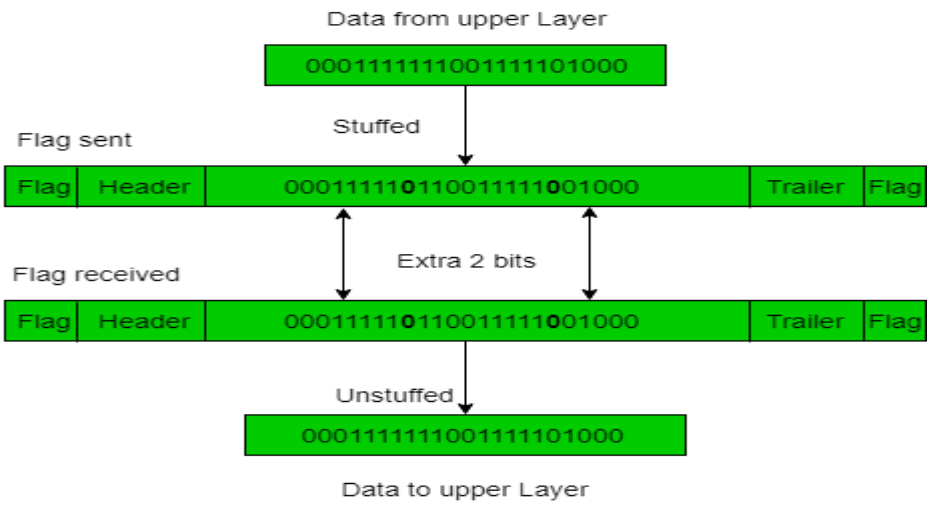
Disadvantage – It is very costly and obsolete method.

3. **Bit Stuffing:** Let ED = 01111 and if data = 01111

   –> Sender stuffs a bit to break the pattern i.e. here appends a 0 in data = 0111**0**1.

   –> Receiver receives the frame.

   –> If data contains 01101, receiver removes the 0 and reads the data.



Examples –

- If Data –> 011100011110 and ED –> 0111 then, find data after bit stuffing?

  –>011**0**10001**1**0**1**1**0**0

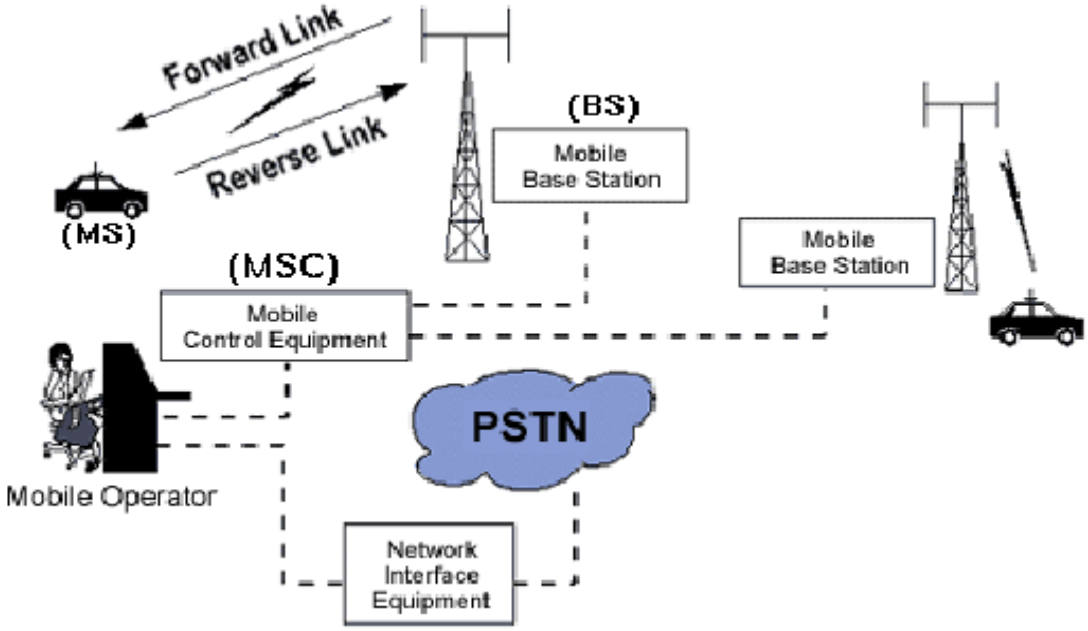
- If Data –> 110001001 and ED –> 1000 then, find data after bit stuffing?

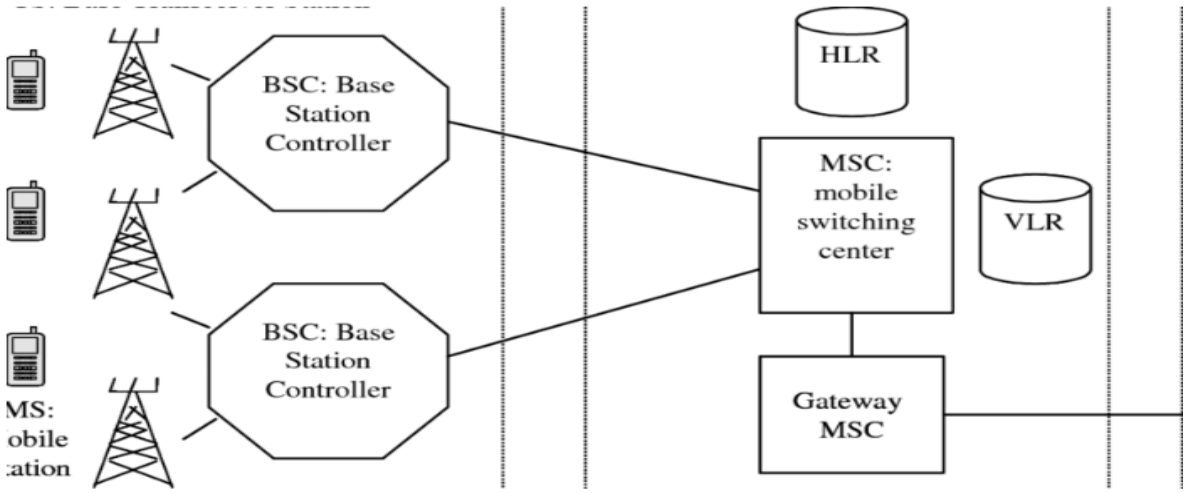
  –> 1100**1**0100**1**1

CELLULAR TELEPHONY

- Cellular telephony is designed to provide communications between two moving units, called mobile stations (MSs), or between one mobile unit and one stationary unit, often called a land unit.
- To make this tracking possible, each cellular service area is divided into small regions called cells. Each cell contains an antenna and is controlled by a solar or AC powered network station, called the base station (BS).

- Each base station, in tum, is controlled by a switching office, called a mobile switching center (MSC). The MSC coordinates communication between all the base stations and the telephone central office.
- It is a computerized center that is responsible for connecting calls, recording call information, and billing





Features of Cellular Systems

- Wireless Cellular Systems solves the problem of spectral congestion and increases user capacity. The features of cellular systems are as follows −

- Offer very high capacity in a limited spectrum.
- Reuse of radio channel in different cells.
- Enable a fixed number of channels to serve an arbitrarily large number of users by reusing the channel throughout the coverage region.
- Communication is always between mobile and base station (not directly between mobiles).
- Each cellular base station is allocated a group of radio channels within a small geographic area called a cell.
- Neighbouring cells are assigned different channel groups.
- By limiting the coverage area to within the boundary of the cell, the channel groups may be reused to cover different cells.
- Keep interference levels within tolerable limits.
- Frequency reuse or frequency planning.
- Organization of Wireless Cellular Network

Shape of Cells

- The coverage area of cellular networks are divided into **cells**, each cell having its own antenna for transmitting the signals.
- Each cell has its own frequencies.
- Data communication in cellular networks is served by its base station transmitter, receiver and its control unit.
- The shape of cells can be either square or hexagon −

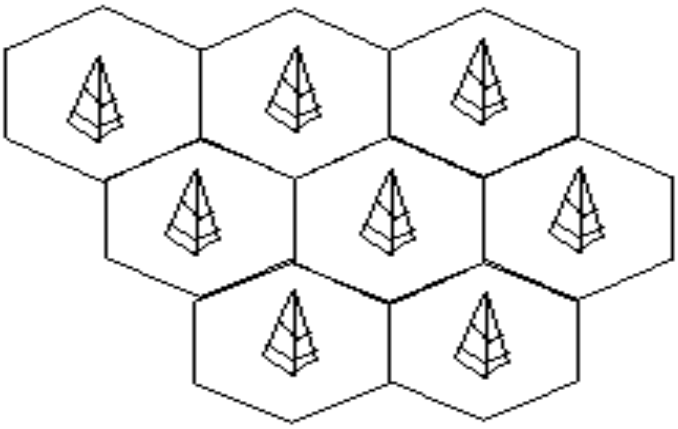
Square

- A square cell has four neighbours at distance **d** and four at distance Root **2 d**
- Better if all adjacent antennas equidistant
- Simplifies choosing and switching to new antenna

Hexagon

- A hexagon cell shape is highly recommended for its easy coverage and calculations. It offers the following advantages −

- Provides equidistant antennas
- Distance from center to vertex equals length of side



Transmitting

- To place a call from a mobile station, the caller enters a code of 7 or 10 digits (a phone number) and presses the send button.
- The mobile station then scans the band, seeking a setup channel with a strong signal, and sends the data (phone number) to the closest base station using that channel.
- The base station relays the data to the MSC.
- The MSC sends the data on to the telephone central office.
- If the called party is available, a connection is made and the result is relayed back to the MSC.
- At this point, the MSC assigns an unused voice channel to the call, and a connection is established.
- The mobile station automatically adjusts its tuning to the new channel, and communication can begin.

Receiving

- When a mobile phone is called, the telephone central office sends the number to the MSC.
- The MSC searches for the location of the mobile station by sending query signals to each cell in a process called *paging.*
- Once the mobile station is found, the MSC transmits a ringing signal and, when the mobile station answers, assigns a voice channel to the call, allowing voice communication to begin.

Roaming

- One feature of cellular telephony is called roaming.
- Roaming means, in principle, that a user can have access to communication or can be reached where there is coverage.
- A service provider usually has limited coverage.
- Neighbouring service providers can provide extended coverage through a roaming contract. The charge for delivery of a letter between two countries can be divided upon agreement by the two countries.

Frequency Reuse

- Frequency reusing is the concept of using the same radio frequencies within a given area that are separated by considerable distance, with minimal interference, to establish communication.

Frequency reuse offers the following benefits −

- Allows communications within cell on a given frequency
- Limits escaping power to adjacent cells
- Allows re-use of frequencies in nearby cells
- Uses same frequency for multiple conversations
- 10 to 50 frequencies per cell

For example, when **N** cells are using the same number of frequencies and **K** be the total number of frequencies used in systems. Then each **cell frequency** is calculated by using the formulae **K/N**.

In Advanced Mobile Phone Services (AMPS) when K = 395 and N = 7, then frequencies per cell on an average will be 395/7 = 56. Here, **cell frequency** is 56.

Handoff Parameters

- The process of transferring a mobile from one channel form one channel or base station to another is called handoff.

- When a mobile moves into a different cell while a conversation is in progress, the MSC automatically transfers the call is to a new channel belonging to the new base station.

Parameters influencing handoff

- Handoff depends on cell size, boundary length, signal strength, fading, reflection and refraction of signals, and manmade noise.
- If the number of channels per area increases if the number of channels allocated per cell is increased or if the area of each cell is decreased.

Handoff initiation

- The base station monitors the signal level for a certain period of time before a handoff is initiated.
- The length of time needed to decide if a handoff is necessary depends on the speed of which the vehicle is moving.

Handoff can be initiated either by the BS or the MS, and it could be due to

1. The radio link
2. Network management
3. Services issues

Radio link type handoff is primarily due to the mobility of the MS and depends on the relative value of the radio link parameters. Radio link type handoff depends on

- Number of MSs that are in the cell
- Number of MSs that have left the cell
- Number of calls transferred to the cell from neighbouring cells by the handoff.
- Number of calls generated in the cell.
- Number and duration of calls terminated in the cell.
- Number of calls handed off to neighbouring cells

- Cell dwell time

Network management may cause handoff if there is a drastic imbalance of traffic over adjacent cells, and optimal balance of channels and other resources are required.

Service related handoff is due to degradation of quality of services (QoS), and handoff could be invoked when such a situation is detected.

The factors that define the right time for handoff are:

- Signal strength
- Signal phase
- Combination of the above two
- Bit error rate (BER)
- Distance

The need for handoff is determined in two different ways:

- Signal strength
- Carrier-to-interference ratio (CIR)

Types of Handoffs

Handoff can be classified into two different types:

- Hard handoff
- Soft handoff

Hard handoff

- Hard handoff, also known as "**break before make**" is one, in which the channel in the source cell is released only and after that the channel in the target cell is engaged.
- Thus the connection to the source is broken before the connection to the target cell is engaged.
- Thus the connection to the source is broken before the connection to the target is made for these reason such handover are also known as break before make. Both FDMA and TDMA employ hard handoff.

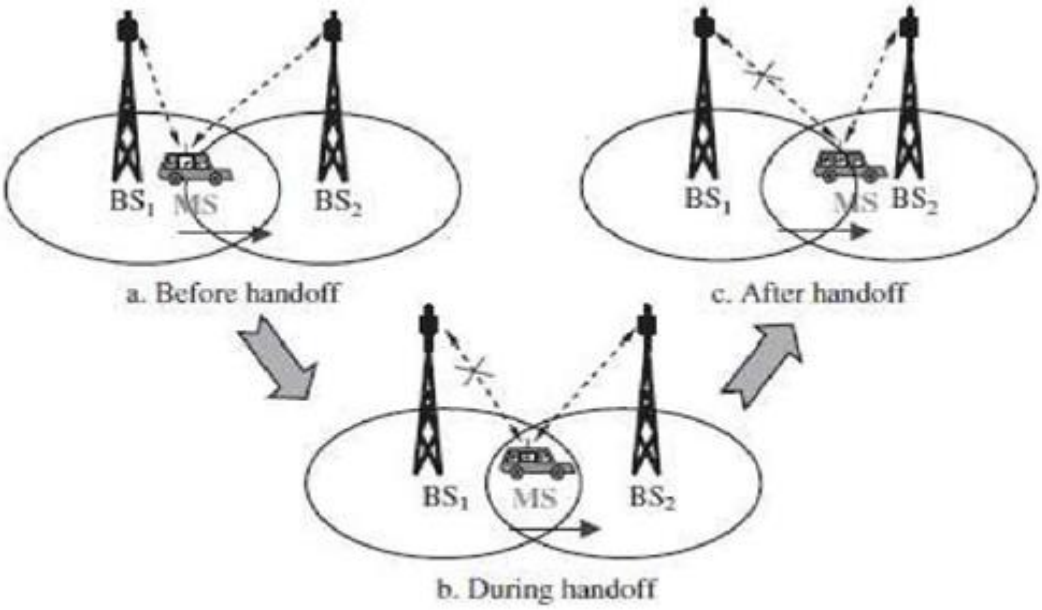


Fig: Hard handoff

Soft handoff

- Soft handoff "**make before break**" is one in which the channel in the source cell is oriented and used for a while in parallel with the channel in the target cell.
- In this case the connection to the target is established before the connection to the source is broken; hence this handover is called soft handoff.

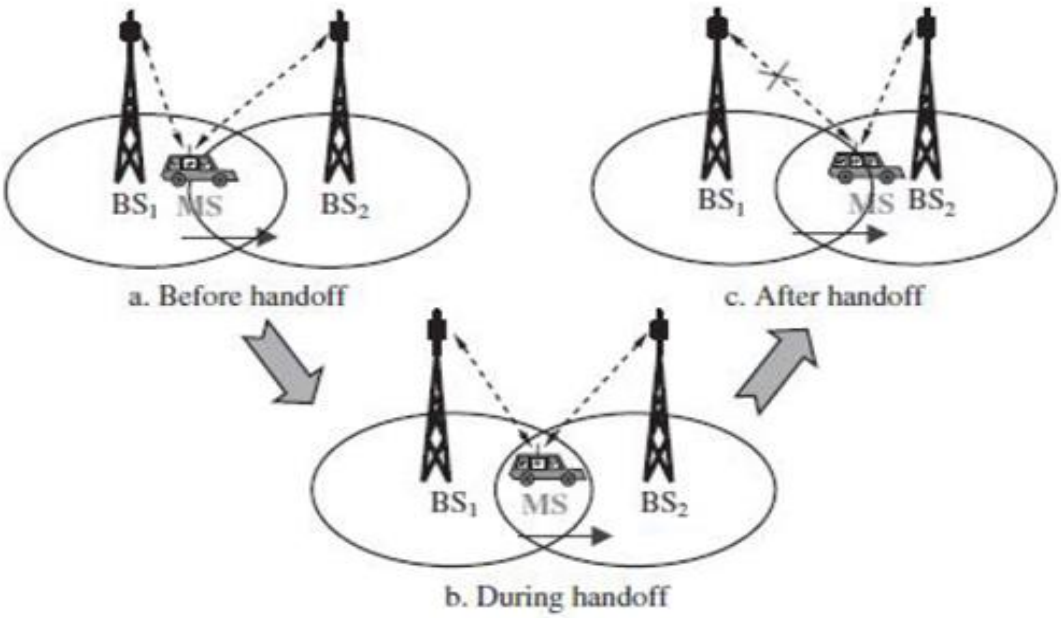


Fig: Soft Handoff

- It is also possible to move from a cell controlled by one MSC area to a cell connected to another MSC.

First Generation Analog cellular phones:

Advanced mobile phone service (AMPS) was developed by Bell Laboratories in the 1970's, and started in the US after FCC allocation in 1983 of 40 MHz paired spectrum in the 800 MHz frequency range. The system used a frequency divided modulation access (FDMA), duplex frequencies for up and down link (frequency division duplexing - FDD), with 30 kHz channels, one user per channel, analog voice modulation (FM), blank and burst transmission.

RF channel	30 kHz
Reuse pattern	typically 7
Duplex	FDD
Multiple access	FDMA
Multiplex	1 traffic channel per RF channel
Voice	FM modulation

Second Generation Digital wireless systems:

Second generation cellular systems are characterized by the introduction of voice digitizing and digital encoding, thus opening a number of DSP possibilities such as forward error correction schemes. Frequency or time division multiple access techniques are used (FDMA or TDMA). Code division multiple access (CDMA) is introduced by Qualcomm (TIA-EIA IS-95, or ANSI-95) and becomes the basis for the main 3G systems. Overall capacity is increased, signaling capabilities and system intelligence is considerably enriched.

RF channel	30 kHz, 200 kHz in GSM, 1.25 MHz for CDMA
Reuse pattern	7 (less with frequency hopping), 1 for CDMA
Duplex	mostly FDD (emergence of TDD)
Multiple access	FDMA, TDMA (8 full-rate time slots for GSM), or CDMA

Voice	Digital encoded: GSM full rate 13.4 kbps, CDMA 13 kbps QCELP or 8 kbps EVRC

Third generation systems:

Digital systems were further improved upon, mostly for higher voice capacity and higher data rates; they evolved into third generation standards.

RF channel	1.25, 5, 10, 15 MHz
Reuse pattern	1 (CDMA)
Duplex	mostly FDD, some TDD
Multiple access	CDMA
Voice	Digital encoded: bit rates 8 kbps and below
Data	Up to several Mbps (3.1 Mbps for EV-DO, 15 Mbps for HSDPA)

Fourth generation systems:

Fourth generation standards deal with higher throughput, low latency, IP network architecture. Air interfaces focus on multicarrier techniques like OFDM, and advanced antenna systems such as multiple input multiple output (MIMO) systems.

RF channel	generally wider: 10, 20 MHz, more
Reuse pattern	1 (OFDMA)
Duplex	FDD or TDD depending on spectrum
Multiple access	OFDMA
Voice	VoIP
Data	IP based, flat architecture, convergence

Fifth generation systems:

Fifth generation definition vary greatly, from software defined radio (SDR) to adaptation to any new radio interface as needed. Today's 5G focus revolve around improving throughput, lowering latency, and target products to be deployed in the year 2020.

Technologies / Features	***1G***	***2G/2.5G***	***3G***	***4G***	***5G***
Evolution	1970	1980	1990	2000	2010
Deployment	1984	1999	2002	2010	2015
Data Rate	2 kbps	14.4-64 kbps	2 Mbps	200 Mbps to 1 Gbps for low mobility	10 Gbps to 100 Gbps
Famous Standards	AMPS	2G: GSM,C]DMA 2.5G: GPRS, EDGE, 1xRTT	WCDMA, CDMA-2000	LTA, WiMAX	Not yet defined
Technology behind	Analog cellular technology	Digital cellular technology	Broad bandwidth CDMA, IP technology	Undefined IP and seamless combination of broadband. LAN/WAN/PAN/ WLAN	Undefined IP and seamless combination of broadband. LAN/WAN/ PAN/WLAN
Service	Voice	2G: Digital Voice, SMS 2.5G: Voice+Data	Integrated high quality audio, video and data	Dynamic information access, wearable devices	Dynamic information access, wearable devices with AI capabilities
Multiplexing	FDMA	TDMA,CDMA	CDMA	CDMA	CDMA
Type of Switching	Circuit	2G: Circuit 2.5G: Circuit and packet	Packet	Packet	Packet
Handoff	Horizontal	Horizontal	Horizontal	Horizontal and Vertical	Horizontal and Vertical
Core Network	PSTN	PSTN	Packet network	Internet	Internet